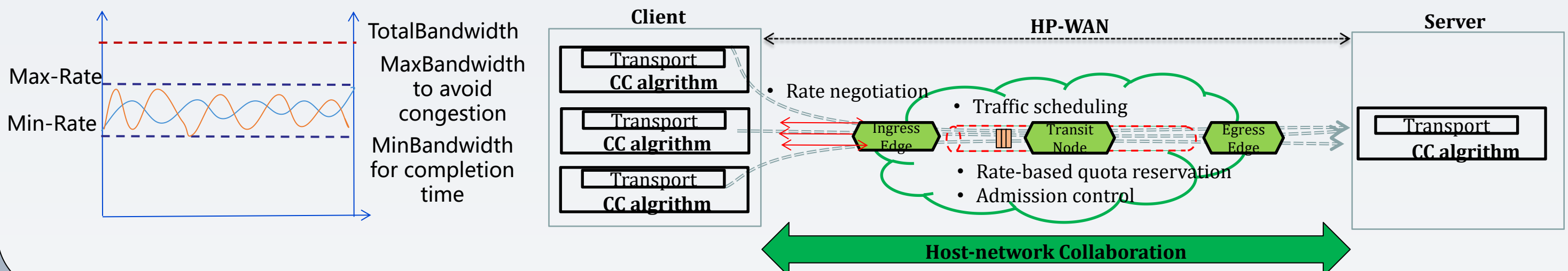




IETF 125 HP-WAN HACKATHON

HP-WAN (High Performance Wide Area Networks)

- A type of Wide Area Network (WAN) designed specifically to meet the *high-speed, low-latency, and high-capacity* needs of scientific research, education, and data-intensive applications.
- The fundamental requirements is high-speed data transmission across WANs which demands *high throughput, low latency, high reliability within a completion time*.
- The existing technologies face the challenges such as *poor convergence speed, unscheduled traffic, long feedback loop and concurrent multi-flows transmission*.
- The framework for HP-WAN enables the *host-and-network collaboration* and particularly *enhances the congestion control* and facilitates the functionalities for the host to collaborate with the network to perform rate negotiation, such as quota-based resource reservation, admission control and traffic scheduling.



HP-WAN Hackathon Project

Champions: Quan Xiong (ZTE), Daniel King (Lancaster University), Kehan Yao (China Mobile), Junfeng Zhao (CAICT)

- ◆ Learn about HPWAN developments in *R&E networks and open-source controllers and orchestrators* for integrated HPWAN connection *performance monitoring* (IETF RestConf/NetConf/YANG).
- ◆ HPWAN integration and *simulation on topologies for public networks* and service scenarios based on the *HP-WAN framework and related functions*, such as the rate negotiation, admission control, traffic scheduling, and resource reservations with distributed signaling (IETF RSVP-TE).
- ◆ Implement the *system optimizations* (e.g., QUIC-based) and protocol modifications for the traffic transmission.
- ◆ Test and *evaluate the performance* (e.g., *throughput, delay, and packet loss*) using RDMA, TCP, and UDP over the existing public network.

Related documents:

- <https://datatracker.ietf.org/doc/draft-kcrh-hpwan-state-of-art/>
- <https://datatracker.ietf.org/doc/draft-xhy-hpwan-framework/>
- <https://datatracker.ietf.org/doc/draft-xiong-hpwan-signaling-solution/>
- <https://datatracker.ietf.org/doc/draft-yx-hpwan-uc-requirements-public-operator/>
- <https://datatracker.ietf.org/doc/draft-zhao-hpwan-scenarios-deployment/>

HPWAN Performance Monitoring

HPWAN requires continuous monitoring because guaranteed or predictable services cannot be delivered on shared infrastructure without real-time visibility of capacity, performance, and faults.

- Monitoring provides the telemetry needed for admission control, rate negotiation, scheduling, and service assurance, allowing orchestrators to make reliable decisions and respond quickly when network conditions change.

A key objective of the HPWAN hackathon at IETF 125 (Shenzhen) is to prototype some collection and reporting for HPWAN services.

- The monitoring stack will need to combine active path measurement, passive traffic analysis, device and interface monitoring, and service-level observability rather than relying on a single platform.
- Capabilities, should include:

1. Active network measurement
2. Infrastructure and node monitoring
3. Passive traffic and flow analysis
4. Service and application observability

- The monitoring data collected for HPWAN will have to include both network and service metrics. At the network level, this includes one-way delay, RTT, jitter, loss, throughput, utilisation, traceroute, path changes, interface status, errors, and flow-level usage. At the service level, it should include reservation success, admitted versus delivered bandwidth, transfer completion time, deadline compliance, controller health, telemetry freshness, and end-to-end workflow success.
- Combined, these measurements provide the visibility needed for admission control, service assurance, troubleshooting, and closed-loop optimization.



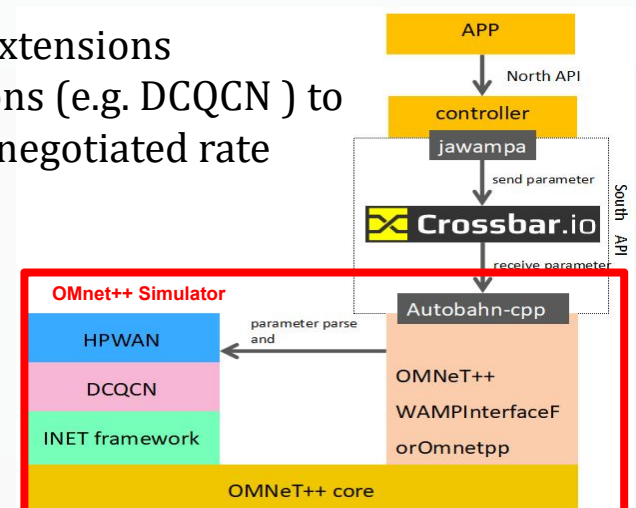
HPWAN OMnet++ Simulator for RSVP-TE Signaling

The framework and functions design for HP-WAN OMnet++ simulator:

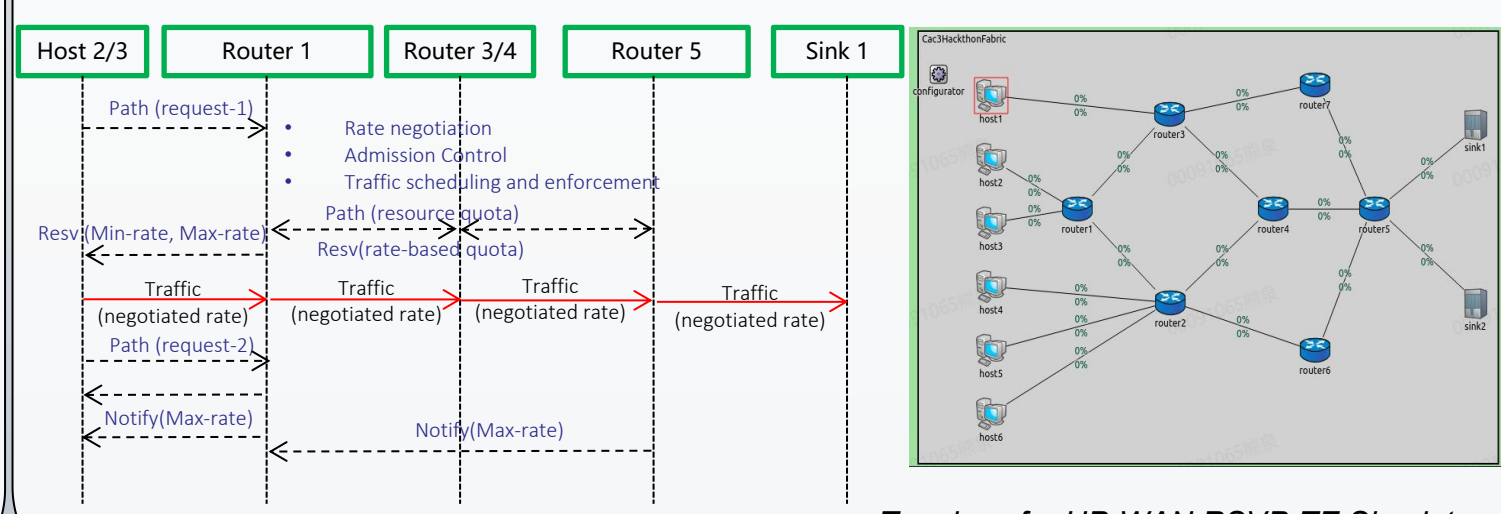
- design the complicated topologies for public networks and service scenarios specially for HP-WAN framework
- implement and simulate the HP-WAN functions such as the rate negotiation, admission control, traffic scheduling and resource reservations with distributed signalling to extend the RSVP-TE libraries to implement the HP-WAN
- signaling solutions through RSVP-TE extensions
- perform the CC algorithms optimizations (e.g. DCQCN) to simulate the traffic transmission with negotiated rate

Simulation Goals:

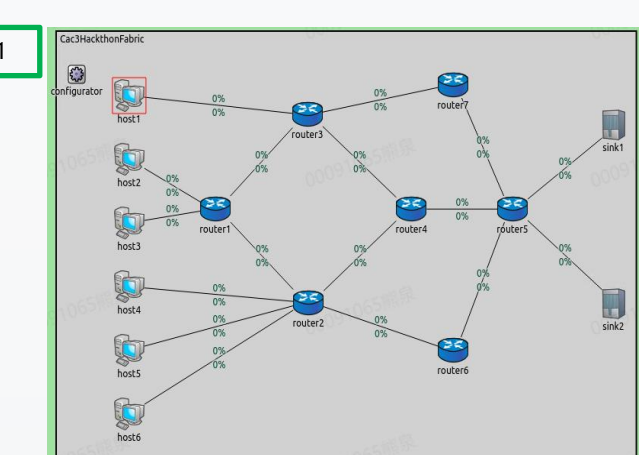
- implement the HP-WAN mechanism with RSVP-TE extensions
- simulate the traffic transmission with HP-WAN framework
- performance comparison between HP-WAN, RSVP-TE and DCQCN



OMnet++ Design for HP-WAN RSVP-TE Simulator

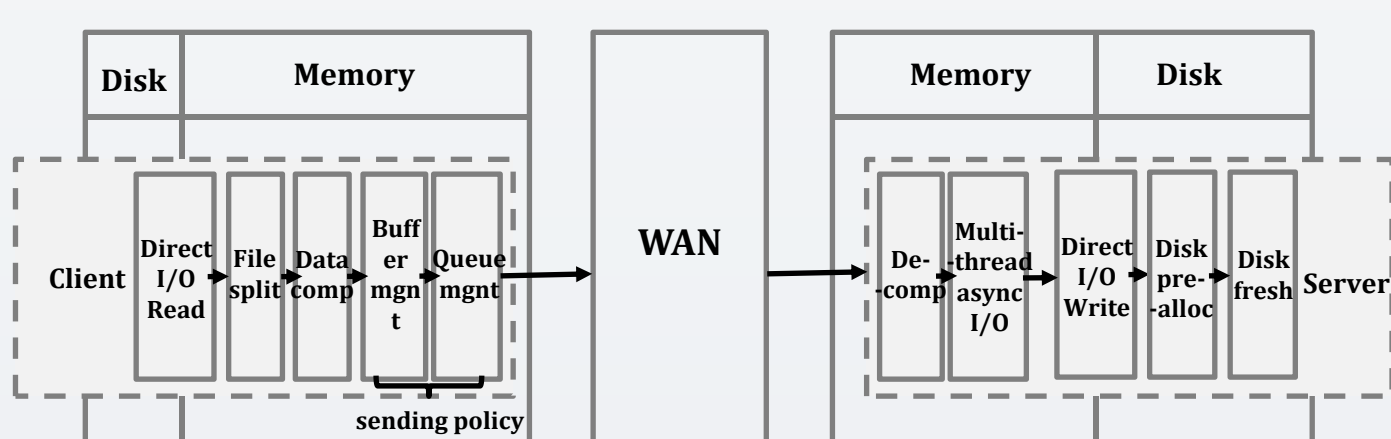


RSVP Extensions for HP-WAN Simulator



Topology for HP-WAN RSVP-TE Simulator

HPWAN QUIC-based System Optimization



System Optimization Overview

QUIC-based Implementations Key Features:

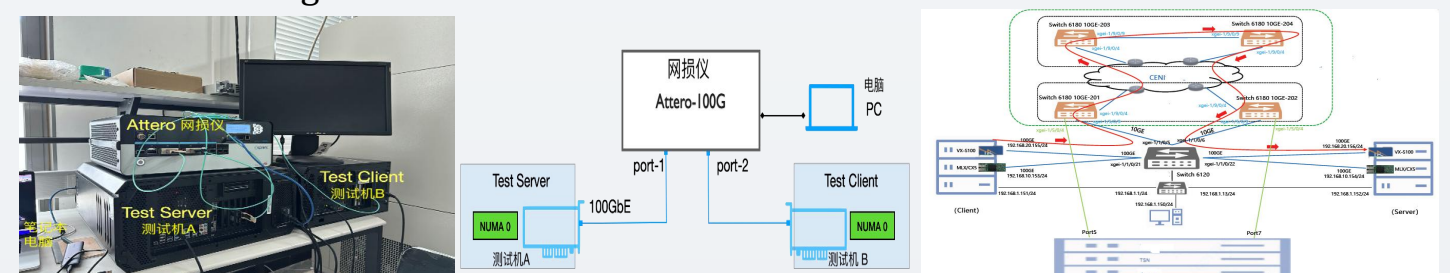
- System optimization for QUIC-based Implementations
 - Direct I/O
 - Disk Pre-allocation
 - Sender-side Transmission Policy Optimization
 - Large File Splitting
 - File Compression
 - Multithreaded Asynchronous I/O
- Protocol modifications over standard QUIC
 - **Encryption:** change TLS 1.3 to optional, to reduce the CPU overhead at end server
 - **QoS:** set priorities to flows that require high-performance transmission
 - **Flow control:** per-stream based buffer management
 - **Congestion control:** use dBBR¹ to replace CUBIC for better fairness

1. To be appeared in IWQoS 26



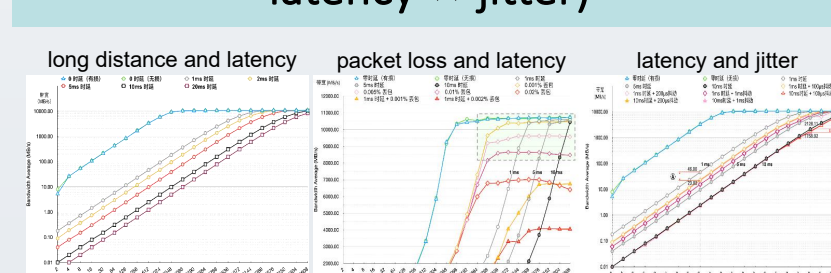
Performance Evaluation of HPWAN: Simulation and Testing

- Introduction for Lab Simulated and trail Testing
- Lab Testing Environment
- Trail Testing topology (RDMA)

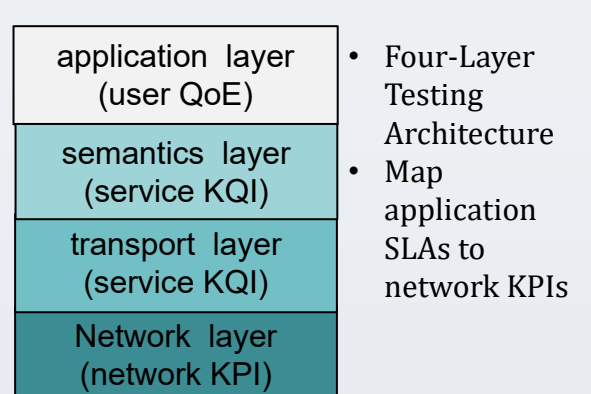


Results and summary :HPWAN Performance Metrics for Wide-Area Network:

High throughput in WAN (RDMA) $\approx N \times \text{message size} / (\rho(\text{packet loss} \times \text{latency} \times \text{jitter}))$



- **Conclusions:**
 - Increasing message size, RDMA achieves >80% throughput efficiency at >2000km distances;
 - Latency makes packet loss requirements exponentially stricter to achieving > 80% throughput requires;
 - Jitter induces more severe throughput degradation than latency.
- **Hierarchical Test and Verification Architecture:**
 - Network layer: L1~L3 Network KPI, latency, jitter, packet loss, throughput, availability, bandwidth utilization
 - Transport layer: throughput, congestion control, flow control
 - Semantics layer: throughput, Memory Read/Write Optimization
 - Application layer: throughput, TTFT, TPS



Thanks for your participating!